
GA
all_data : dict
bcm : Union[BCModel, ase.Atoms]
composition : ndarray
continued : int
dt_created : datetime
e : int
formula : str
generation : int
has_run : bool
is_monometallic
last_min : int
max_gens : int
mute_pct
n_mute
n_mute_atomswaps : Optional[int]
num_atoms
orig_min
pop : list
popsize : int
prev_results : NoneType
random : bool
runtime : int
save_every : int
shape : str
spike : bool
stats : list, ndarray
use_metropolis : bool
continue_run(max_gens: int, max_nochange: int, min_gens: int)
initialize_new_run()
is_new_min(check_db: bool): bool
make_atoms_object(np_index: int): ase.Atoms
plot_results(save_path: str, ax: plt.Axes, show: bool): Union[plt.Figure, plt.Axes]
run(max_gens: int, max_nochange: int, min_gens: int)
save_ga_pickle(path: str): str
save_to_db()
sort_pop()
summ_results(display: bool): str
view_np(np_index: int)
Nanoparticle
bcm
ce
composition : ndarray
num_atoms
ordering : ndarray
copy(): Nanoparticle
mate(nanop2: Nanoparticle): List[Nanoparticle]
mutate(n_swaps: int)
bcm
bcm
AtomGraph
cns : ndarray
coeffs : dict
max_cn
min_cn
mono_ce0
mono_ce1
num_atoms
symbols : tuple
unique_cns
calcMixing(ordering, holder_array)
calc_ce(ordering)
calc_cn_dist(ordering)
calc_ee(ordering)
countMixing(ordering, holder_array)
getLocalCE(ordering, index)
get_adjacency_list()
metropolis(ordering, num_steps, swap_any)
set_composition(kind0, kind1)
Atoms
id
index
nanoparticle
structure_id
x
y
z
BCModel
CN_Method : str
a1
a2
atoms
avg_radius
bond_list : Optional[Iterable], list, ndarray
ce_bulk : NoneType, dict
cn : ndarray
cn_precomps : NoneType
gammas : NoneType, dict
info : dict
inv_radii : ndarray
metal_types
num_shells
precomps : NoneType, ndarray
radii_bond_list
radius : dict
shell_map
syms
calc_ce(orderings: np.ndarray): float
calc_ee(orderings: np.ndarray): float
calc_gmix(orderings: np.ndarray, T: float): float
calc_smix(orderings: np.ndarray): float
get_info()
metropolis(ordering: np.ndarray, num_steps: int): None
BimetallicLog
batch_run_num
date
ga_generations
id
metal1
metal2
new_min_structs
runtime
shape
shell_range
tot_structs
BimetallicResults
CE
EE
atoms_obj : NoneType
compressed_ordering
diameter
id
last_updated
metal1
metal2
n_metal1
n_metal2
num_atoms
ordering
shape
smix : NoneType
structure_id
build_atoms_obj()
build_central_rdf(nbins, pcty)
build_chem_formula(latex, bold)
build_prdf(alpha, beta, dr)
build_prdf_plot()
get_atoms_obj()
get_chemical_formula(latex, bold)
get_gmix(T)
save_np(path)
show()
GAError
GammaValues
bde_aa
bde_ab
bde_bb
ce_a
ce_b
cn_max
cnbulk_a
cnbulk_b
datafile_ce
datafile_cn
datafile_experimental_hbe
datafile_theoretical_hbe
element_a
element_b
gamma : dict
calc_coeffs_dict(max_cn)
calculate_gamma(element1, element2)
calculate_total_gamma(cn, element1, element2)
lookup_bde(a, b)
lookup_bulk_coordination(element)
lookup_cohesive_energy(element)
NPBuilder
cuboctahedron(num_shells, kind)
elongated_pentagonal_bipyramid(num_shells, kind)
fcc_cube(num_units, kind)
icosahedron(num_shells, kind)
sphere(num_layers, kind, unit_cell_length)
NPDatabaseError
Nanoparticles
atoms
atoms_obj : NoneType
bimetallic_results
bonds_list : NoneType
id
num_atoms
num_bonds : NoneType
num_shells
polymetallic_results
shape
get_atoms_obj()
get_atoms_obj_skel()
get_bonds_list()
get_diameter()
load_bonds_list()
PolymetallicResults
CE
EE
atoms_obj : NoneType
composition : NoneType
composition_list
id
last_updated
metals : NoneType
metals_list
num_atoms
ordering : NoneType
ordering_string
shape
smix
structure_id
get_chemical_formula(latex, bold)
get_gmix(T)
save_np(path)
show()